

Distributed Systems

Unit VI: Recent Trends in Distributed Systems

Syllabus:

Recent Trends: Introduction, Portable and handheld Devices, Wearable devices, Devices embedded in appliances, Parallel Virtual Machine (PVM), Jini, Service Oriented Architecture, The Future of Recent Trends.

Tools for Distributed System Monitoring: Prometheus, Zabbix, Nagios

Case Studies: Mach, Chorus

◆ Recent Trends

❖ Introduction

❖ Portable and handheld Devices

Portable and Handheld Devices in Recent Trends of Distributed Systems

Portable and handheld devices are one of the most important recent trends in distributed systems. These devices are small, mobile computing systems that can connect to distributed networks anytime and anywhere using wireless communication technologies such as Wi-Fi, Bluetooth, 4G/5G, and the Internet.

Definition

Portable and handheld devices are lightweight computing devices that support mobility and communication in distributed environments.

Examples include:

- Smartphones
- Tablets
- Laptops
- Smartwatches
- Wearable devices
- PDAs

- IoT devices and smart sensors

Characteristics

1. Mobility

Users can access distributed services while moving from one location to another.

2. Wireless Connectivity

Devices communicate through wireless networks like:

- Wi-Fi
- Bluetooth
- Cellular networks
- Satellite communication

3. Small Size and Portability

These devices are compact, lightweight, and easy to carry.

4. Resource Constraints

Compared to servers, handheld devices have:

- Limited battery power
- Less memory
- Lower processing capability

5. Ubiquitous Access

Users can access services anytime and anywhere. This is called **ubiquitous computing**.

Role in Distributed Systems

Portable devices act as:

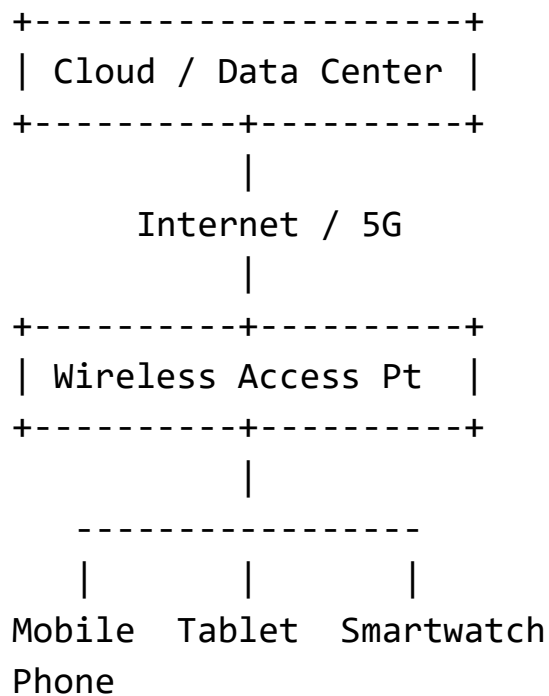
- **Clients** accessing remote servers
- **Data collection nodes** in IoT systems
- **Edge devices** processing data locally

- **Communication endpoints** in cloud services

They help distributed systems become:

- More flexible
- More scalable
- Location independent

Architecture of Portable Devices in Distributed Systems



Applications

1. Mobile Banking

Users perform banking transactions through distributed banking servers.

2. E-Commerce

Portable devices access distributed online shopping systems.

3. Healthcare Systems

Wearable devices continuously monitor patient health and send data to distributed hospital systems.

4. Smart Homes

Mobile phones control smart appliances connected through IoT networks.

5. Multimedia Streaming

Users watch distributed multimedia services like video streaming and video conferencing.

6. GPS and Navigation

Location data is processed using distributed mapping services.

Advantages

- Access services from anywhere
- Real-time communication
- Improved productivity
- Supports remote work and learning
- Easy integration with cloud computing
- Enables IoT and smart environments

Challenges

1. Security Issues

Wireless communication is vulnerable to:

- Hacking
- Data theft
- Unauthorized access

2. Limited Battery Power

Continuous communication consumes battery quickly.

3. Network Dependency

Performance depends on network availability and signal strength.

4. Device Heterogeneity

Different operating systems and hardware make interoperability difficult.

5. Data Synchronization

Maintaining consistency across distributed devices is challenging.

Recent Trends

Edge Computing

Processing is moved closer to portable devices to reduce latency.

AI on Devices

Modern smartphones and wearables now include AI processors (NPUs) for distributed AI computing.

Internet of Things (IoT)

Portable and embedded devices are increasingly interconnected in smart cities, healthcare, and industrial systems.

5G Technology

5G improves:

- Speed
- Reliability
- Real-time communication

for mobile distributed applications.

Wearable Computing

Smart glasses, smartwatches, and fitness trackers are becoming integral parts of distributed systems.

Conclusion

Portable and handheld devices have transformed distributed systems by enabling mobile and ubiquitous computing. They allow users to access distributed resources anytime and anywhere, making modern applications such as cloud computing, IoT, smart healthcare, and multimedia streaming possible. Despite challenges like security and limited resources, advances in

wireless communication, edge computing, AI, and 5G continue to expand their role in distributed environments.

❖ **Wearable devices**

May_Jun_2024

Q7) a) How wearable devices work in distributed systems? Discuss the problems involved with wearable computing.

Ans. Wearable Devices in Distributed Systems

Wearable devices are small electronic devices that can be worn on the body and are connected to distributed systems through wireless networks. They collect, process, and exchange data continuously with nearby devices, cloud servers, and IoT systems.

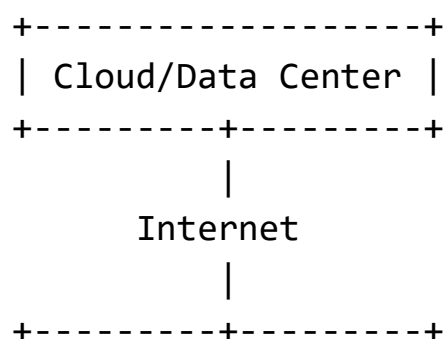
Examples:

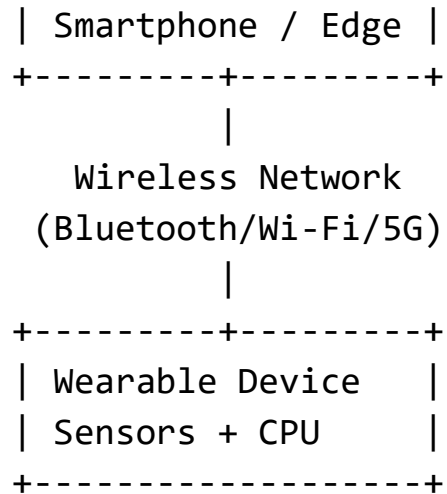
- Smartwatches
- Fitness bands
- Smart glasses
- Health monitoring devices
- AR/VR headsets

Working of Wearable Devices in Distributed Systems

Wearable devices operate as distributed computing nodes that communicate with smartphones, edge devices, and cloud servers.

Basic Working Architecture





Step-by-Step Working

1. Data Collection

Wearable devices contain sensors that collect real-time data such as:

- Heart rate
- Body temperature
- Movement
- GPS location
- Blood oxygen level

2. Local Processing

The wearable performs basic computation using embedded processors:

- Filtering sensor data
- Detecting events
- Compressing information

3. Wireless Communication

Data is transmitted through:

- Bluetooth
- Wi-Fi
- NFC

- 4G/5G

Usually, the wearable sends data to a smartphone or gateway device.

4. Edge or Cloud Processing

The smartphone or cloud server performs advanced processing:

- AI-based analysis
- Long-term storage
- Pattern recognition
- Health monitoring

5. Feedback to User

Results are returned to the wearable or mobile app:

- Notifications
- Health alerts
- Fitness reports
- Navigation assistance

Role of Wearable Devices in Distributed Systems

Wearables act as:

- Mobile clients
- IoT nodes
- Sensor devices
- Edge computing components

They support:

- Ubiquitous computing
- Real-time monitoring
- Distributed healthcare
- Smart environments

Applications of Wearable Computing

1. Healthcare Systems

Continuous patient monitoring using smartwatches and medical wearables.

Example:

- ECG monitoring
- Blood pressure tracking
- Emergency alerts

2. Fitness Tracking

Fitness bands collect and synchronize activity data with cloud servers.

3. Smart Navigation

Smart glasses provide real-time navigation and location services.

4. Industrial Safety

Workers wear smart helmets or sensors for safety monitoring in industries.

5. Augmented Reality (AR)

AR headsets interact with distributed cloud systems for real-time rendering and collaboration.

Problems Involved with Wearable Computing

1. Limited Battery Power

Wearable devices are very small, so battery capacity is limited.

Problems:

- Frequent charging
- Reduced performance
- Continuous sensing drains power

2. Security and Privacy Issues

Wearables collect sensitive personal data.

Risks include:

- Health data leakage
- Unauthorized tracking
- Data theft
- Identity misuse

3. Limited Processing Capability

Wearables have:

- Small CPUs
- Limited RAM
- Limited storage

Hence they depend heavily on cloud or edge servers.

4. Connectivity Problems

Wireless communication may suffer from:

- Signal loss
- Interference
- Network congestion
- High latency

5. Data Synchronization

Maintaining consistency between:

- Wearable
- Smartphone
- Cloud database

is difficult in distributed environments.

6. Device Heterogeneity

Different devices use:

- Different operating systems
- Different protocols

- Different hardware architectures

This creates interoperability issues.

7. Scalability Issues

Large numbers of wearable devices generate massive real-time data, increasing:

- Network traffic
- Storage requirements
- Processing load

8. Mobility Management

Users move continuously between networks and locations.

Distributed systems must support:

- Seamless handoff
- Continuous connectivity
- Location management

9. User Interface Limitations

Wearables have:

- Small screens
- Limited input methods

This restricts interaction and usability.

10. Reliability Issues

Sensor failures or communication failures can affect:

- Accuracy
- Availability
- Real-time monitoring

Especially critical in healthcare applications.

Advantages of Wearable Computing

- Continuous monitoring
- Real-time communication
- Improved healthcare services
- Hands-free operation
- Mobility and convenience
- Integration with IoT and cloud computing

Recent Trends in Wearable Computing

Edge Computing

Processing data near the wearable reduces latency and bandwidth usage.

AI-Based Wearables

Modern wearables use AI for:

- Health prediction
- Activity recognition
- Personalized recommendations

5G Integration

5G improves:

- Faster communication
- Low latency
- Reliable connectivity

Smart Healthcare Ecosystems

Wearables are increasingly integrated with hospital information systems and remote healthcare platforms.

Conclusion

Wearable devices are important components of modern distributed systems. They enable continuous sensing, mobility, real-time communication, and

smart services through integration with cloud computing, IoT, and wireless networks. However, wearable computing faces several challenges such as limited battery power, security concerns, connectivity issues, and resource constraints. Advances in AI, edge computing, and 5G technologies are helping overcome these problems and expanding the role of wearable systems in everyday life.

❖ **Devices embedded in appliances**

Devices Embedded in Appliances in Distributed Systems

Devices embedded in appliances are small computing systems integrated into household, industrial, or commercial appliances to provide automation, communication, monitoring, and intelligent control. These embedded devices are important components of modern distributed systems and the Internet of Things (IoT).

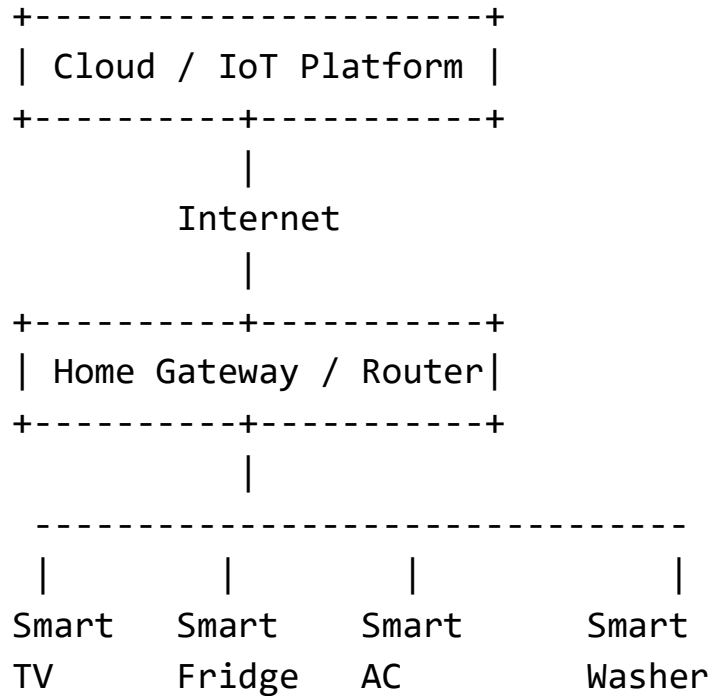
Examples:

- Smart TVs
- Smart refrigerators
- Washing machines
- Microwave ovens
- Air conditioners
- Smart lighting systems
- Industrial machines
- Smart security systems

Definition

An embedded device is a specialized computing unit designed to perform dedicated functions within an appliance. In distributed systems, these devices communicate with other devices, servers, and cloud platforms through networks.

Architecture of Embedded Appliances in Distributed Systems



Components of Embedded Devices

1. Microprocessor / Microcontroller

Controls appliance operations.

2. Sensors

Collect information such as:

- Temperature
- Motion
- Humidity
- Pressure

3. Memory

Stores programs and data.

4. Communication Module

Provides connectivity using:

- Wi-Fi
- Bluetooth
- ZigBee
- Ethernet

5. Embedded Software

Controls appliance behavior and communication.

Working of Embedded Appliances in Distributed Systems

1. Data Collection

Sensors collect information from the environment.

Example:

A smart AC senses room temperature.

2. Local Processing

Embedded processors analyze sensor data and make decisions.

Example:

The AC automatically adjusts cooling.

3. Communication with Other Devices

The appliance exchanges data with:

- Smartphones
- Home gateways
- Cloud servers

4. Remote Monitoring and Control

Users control appliances remotely using mobile apps or web services.

Example:

Turning on a washing machine from a smartphone.

5. Cloud Integration

Cloud platforms provide:

- Data storage
- Analytics
- AI-based automation
- Device management

Role in Distributed Systems

Embedded devices act as:

- Distributed nodes
- IoT endpoints
- Smart sensors
- Edge computing devices

They contribute to:

- Automation
- Resource sharing
- Real-time monitoring
- Intelligent decision making

Applications

1. Smart Homes

Home appliances coordinate through distributed networks.

Examples:

- Smart lighting
- Smart thermostats
- Voice-controlled devices

2. Healthcare Equipment

Embedded medical devices monitor patient conditions and communicate with hospital servers.

3. Industrial Automation

Machines with embedded controllers support distributed manufacturing systems.

Examples:

- Robotic systems
- Smart sensors
- Automated production lines

4. Smart Energy Systems

Smart meters and appliances optimize power consumption.

5. Smart Security Systems

Embedded cameras and alarms communicate with monitoring systems.

Advantages

1. Automation

Reduces human effort through automatic operations.

2. Real-Time Monitoring

Continuously tracks system status and environment conditions.

3. Remote Accessibility

Users can control appliances from anywhere.

4. Energy Efficiency

Smart devices optimize resource utilization.

5. Improved Reliability

Distributed monitoring improves fault detection and maintenance.

Problems and Challenges

1. Security Issues

Embedded appliances are vulnerable to:

- Hacking
- Malware
- Unauthorized access

Example:

Attackers gaining control of smart home devices.

2. Privacy Concerns

Devices may collect sensitive user information.

Examples:

- Usage patterns
- Audio/video recordings
- Location data

3. Resource Constraints

Embedded systems often have:

- Limited memory
- Limited CPU power
- Limited battery life

4. Network Dependency

Appliances rely on stable internet connectivity.

Failures may reduce functionality.

5. Device Heterogeneity

Different manufacturers use different:

- Protocols
- Operating systems

- Communication standards

This creates interoperability problems.

6. Scalability Issues

Managing large numbers of embedded devices becomes difficult in large distributed environments.

7. Maintenance and Updates

Firmware updates and bug fixes must be managed remotely and securely.

Recent Trends

Internet of Things (IoT)

Embedded appliances are increasingly interconnected through IoT ecosystems.

Edge Computing

Processing is shifted closer to appliances for faster response times.

AI-Enabled Smart Devices

Modern appliances use AI for:

- Predictive maintenance
- Voice recognition
- Smart recommendations

5G Connectivity

5G improves:

- Low latency
- Faster communication
- Reliable connectivity

Comparison: Traditional vs Embedded Smart Appliances

Feature	Traditional Appliance	Embedded Smart Appliance
Connectivity	No	Yes
Automation	Limited	High
Remote Control	Not possible	Possible
Intelligence	Fixed functions	Adaptive and programmable
Monitoring	Manual	Real-time

Conclusion

Devices embedded in appliances are essential components of modern distributed systems. They combine embedded computing, networking, sensors, and cloud technologies to create intelligent and interconnected environments. These devices enable automation, remote control, real-time monitoring, and smart decision-making in homes, industries, healthcare, and energy systems. Despite challenges such as security, interoperability, and limited resources, advancements in IoT, AI, edge computing, and 5G continue to enhance embedded appliance technologies.

❖ **Parallel Virtual Machine (PVM)**

❖ **Jini**

Nov_Dec_2024

Q7) a) Explain the following in brief: i) Wearable devices ii) PVM iii) JINI

Ans. i) PVM (Parallel Virtual Machine)

Definition

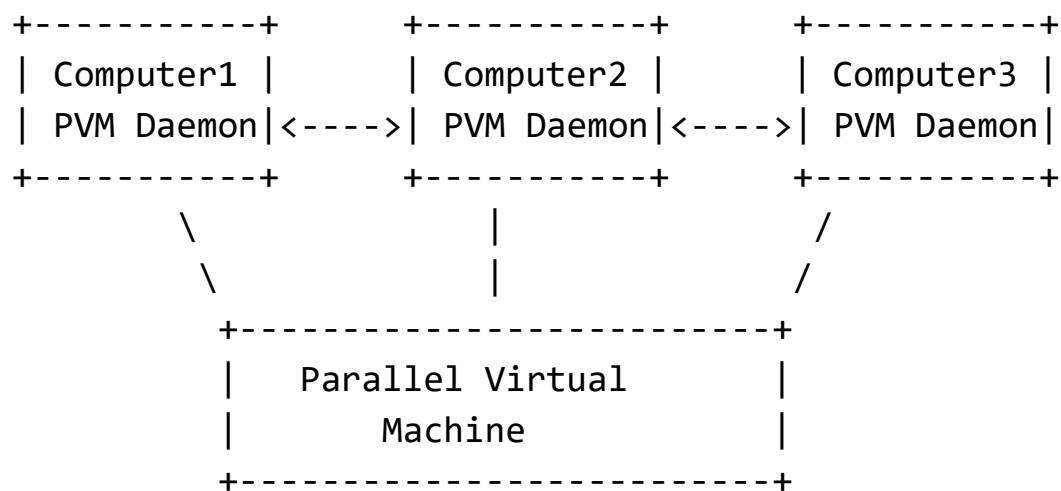
PVM (Parallel Virtual Machine) is a software system that allows a collection of heterogeneous computers connected through a network to work together as a single parallel distributed computing system.

It was developed to support distributed and parallel processing on networked computers.

Features of PVM

- Supports heterogeneous systems
- Allows parallel execution of tasks
- Uses message passing for communication
- Fault tolerant in distributed environments
- Dynamic addition/removal of hosts
- Supports distributed computing clusters

Architecture of PVM



Each computer runs a **PVM daemon** (pvmd) that manages communication and task coordination.

Components of PVM

1. PVM Daemon (pvmd)

- Runs on each host
- Controls communication
- Manages tasks

2. PVM Library

Provides APIs for:

- Message passing
- Task creation
- Synchronization

3. Host Pool

A collection of computers participating in computation.

Working of PVM

Step 1: Host Configuration

Multiple computers are connected in a network.

Step 2: Starting PVM Daemons

Each host runs the pvmd daemon.

Step 3: Task Distribution

Large problems are divided into smaller subtasks.

Step 4: Parallel Execution

Tasks execute simultaneously on multiple machines.

Step 5: Message Passing

Processes communicate through send/receive messages.

Step 6: Result Collection

Partial results are combined to produce the final output.

Communication in PVM

PVM uses:

- Point-to-point communication
- Group communication
- Message passing primitives

Important functions:

- `pvm_send()`

- `pvm_recv()`
- `pvm_spawn()`

Advantages of PVM

- Cost-effective parallel computing
- Supports heterogeneous systems
- Scalable architecture
- Efficient resource utilization
- Flexible and portable

Disadvantages of PVM

- Communication overhead
- Complex programming
- Less efficient than tightly coupled supercomputers
- Security limitations

Applications of PVM

- Scientific simulations
- Weather forecasting
- Image processing
- Distributed research computing
- Engineering analysis

ii) JINI

Definition

JINI is a distributed computing technology developed by Sun Microsystems for building dynamic distributed systems in Java.

It enables devices and services to join or leave a network automatically and interact with each other through service discovery.

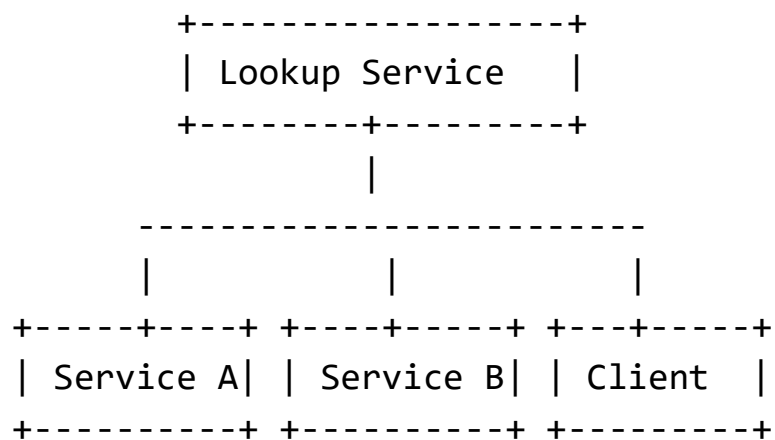
JINI is based on:

- Java technology
- Remote Method Invocation (RMI)
- Distributed object systems

Features of JINI

- Dynamic service discovery
- Plug-and-play networking
- Distributed service architecture
- Platform independence
- Fault tolerance
- Easy service integration

Architecture of JINI



Components of JINI

1. Lookup Service

Acts as a directory for available services.

Functions:

- Service registration
- Service discovery

2. Service Provider

Provides services to clients.

Examples:

- Printer service
- File service
- Database service

3. Client

Searches and uses available services.

4. Java RMI

Supports remote communication between distributed objects.

Working of JINI

Step 1: Service Registration

A service registers itself with the lookup service.

Step 2: Service Discovery

Clients query the lookup service for required services.

Step 3: Downloading Service Proxy

The client downloads a Java proxy object.

Step 4: Remote Interaction

The client communicates with the service using the proxy.

Important Concepts in JINI

1. Leasing

Services register for a limited time period.

If lease expires, the service is removed automatically.

2. Distributed Events

Supports event notification among distributed components.

3. Transactions

Ensures consistency in distributed operations.

Advantages of JINI

- Dynamic networking
- Automatic service discovery
- Platform independence
- Simplifies distributed application development
- Supports mobile and embedded devices

Disadvantages of JINI

- Java dependent
- Security complexity
- Higher memory usage
- Less popular in modern systems

Applications of JINI

- Distributed office systems
- Smart device networks
- Network printers
- Home automation
- Distributed enterprise applications

Difference Between PVM and JINI

Feature	PVM	JINI
Purpose	Parallel computing	Distributed service management
Communication	Message passing	Java RMI
Language Support	C, C++, Fortran	Java
Main Goal	High-performance computing	Dynamic service discovery
Environment	Cluster computing	Network services
Architecture	Parallel virtual machine	Service-oriented architecture

Conclusion

PVM and JINI are important distributed computing technologies designed for different purposes. PVM focuses on parallel computation by combining multiple computers into a virtual parallel machine, while JINI focuses on dynamic service discovery and interaction in distributed Java environments. Both technologies contributed significantly to the development of modern distributed systems and distributed computing models.

❖ Service Oriented Architecture

May_Jun_2023

Q7) b) Explain with an application (e.g., Travel Booking Service) various key components of Service Oriented Architecture

Ans. Service Oriented Architecture (SOA)

Definition

Service Oriented Architecture (SOA) is a software architecture in which applications are built using reusable and loosely coupled services that communicate over a network.

Each service performs a specific business function and can be accessed independently by other applications.

Examples of services:

- Payment service
- Booking service
- Authentication service
- Notification service

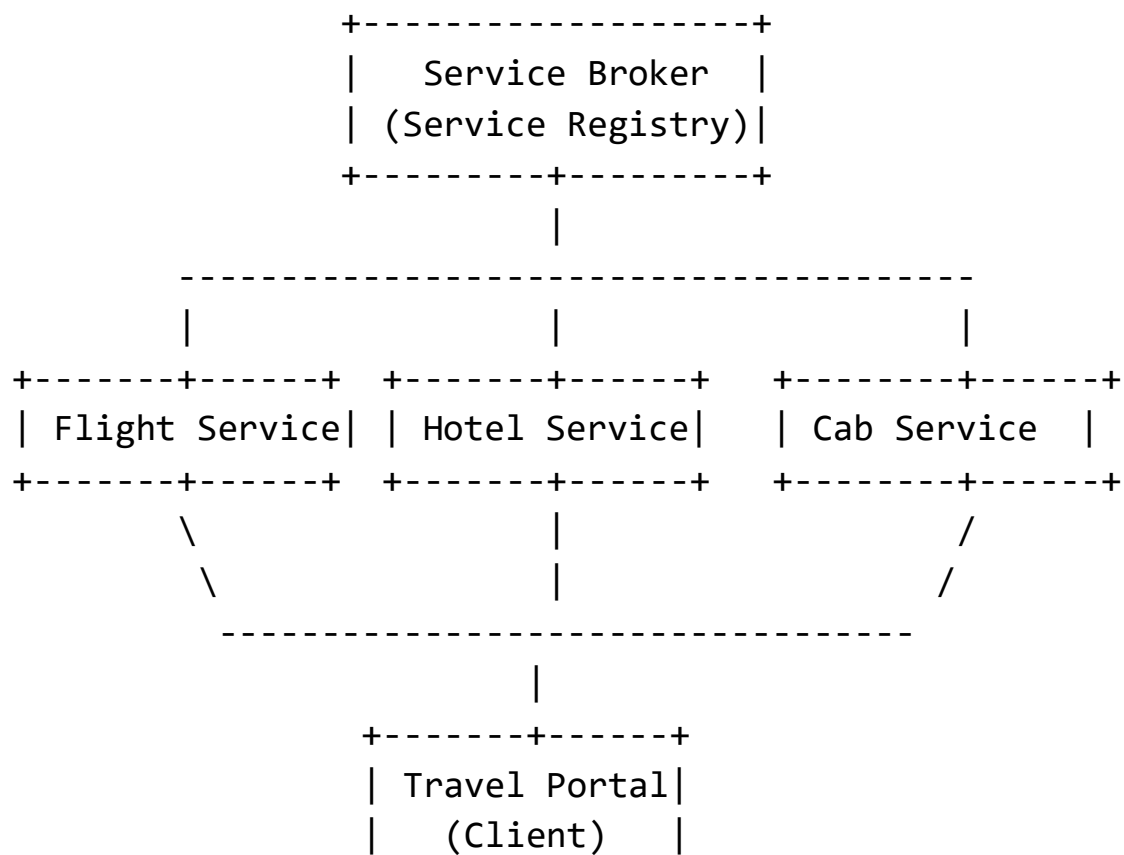
Key Components of SOA Using Travel Booking Service

Consider an online travel booking system where a user books:

- Flights
- Hotels
- Cabs

through different distributed services.

SOA Architecture for Travel Booking Service



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Key Components of SOA

1. Service Provider

Definition

The service provider creates and offers services to clients.

In the travel booking application:

- Airline companies provide flight booking services
- Hotels provide room booking services
- Cab companies provide transportation services

Responsibilities

- Publish services
- Process client requests
- Maintain business logic

Example

Flight Service Provider offers:

- Flight search
- Seat reservation
- Ticket booking

2. Service Consumer (Client)

Definition

The service consumer requests and uses services.

In the travel booking system:

- The traveler using the travel portal acts as the service consumer.

Responsibilities

- Discover services
- Send requests
- Receive responses

Example

A user searches for:

- Available flights
- Hotel rooms
- Cab availability

through the travel application.

3. Service Registry / Broker

Definition

A service registry stores information about available services.

It acts as a directory where providers publish services and consumers discover them.

Responsibilities

- Service registration
- Service lookup
- Service discovery

Example

The travel portal queries the registry to find:

- Airline services
- Hotel services
- Payment gateways

4. Service Contract

Definition

A service contract defines:

- Service interface
- Input/output formats
- Communication rules

Usually described using:

- WSDL (Web Services Description Language)

Example

Flight booking service contract specifies:

- Passenger details
- Flight information
- Ticket format

5. Communication Protocol

Definition

Services communicate through standard protocols.

Common protocols:

- HTTP
- SOAP
- REST
- XML
- JSON

Example

The travel portal sends booking requests using:

- REST APIs
- SOAP messages

6. Business Process / Service Composition

Definition

Multiple services are combined to complete a business task.

Example in Travel Booking

Booking a trip may involve:

1. Flight reservation
2. Hotel booking
3. Cab reservation
4. Payment processing
5. Notification service

All these services work together to complete one travel transaction.

7. Enterprise Service Bus (ESB)

Definition

ESB acts as middleware for communication among services.

Functions:

- Message routing
- Data transformation
- Protocol conversion

Example

The ESB routes:

- Booking requests
- Payment confirmations
- Notification messages

between services.

8. Security Service

Definition

Provides authentication and authorization.

Example

The travel portal verifies:

- User login
- Payment authorization
- Secure transactions

using encryption and authentication mechanisms.

9. Database / Data Service

Definition

Stores and manages distributed application data.

Example

Databases store:

- Flight schedules
- Hotel availability
- Customer records
- Booking history

Working of Travel Booking SOA Application

Step 1: User Request

The user searches for travel options.

Step 2: Service Discovery

Travel portal discovers required services through the registry.

Step 3: Service Invocation

Portal invokes:

- Flight service
- Hotel service
- Cab service

Step 4: Data Exchange

Services exchange information using standard protocols.

Step 5: Payment Processing

Payment service processes transaction securely.

Step 6: Confirmation

Booking confirmation is sent to the user.

Characteristics of SOA

1. Loose Coupling

Services are independent.

2. Reusability

Services can be reused in multiple applications.

3. Interoperability

Different platforms can communicate.

4. Scalability

Services can be scaled independently.

5. Platform Independence

Works across heterogeneous systems.

Advantages of SOA

- Easy integration
- Service reuse
- Reduced development cost
- Better scalability

- Flexibility
- Simplified maintenance

Challenges of SOA

1. Security Issues

Distributed services increase security risks.

2. Performance Overhead

Network communication may increase latency.

3. Complex Service Management

Managing many services becomes difficult.

4. Reliability Problems

Failure of one service can affect the system.

Real-World Examples of SOA

- Online travel booking systems
- Banking systems
- E-commerce platforms
- Healthcare systems
- Cloud computing platforms

Conclusion

Service Oriented Architecture (SOA) enables distributed applications to be built using reusable and interoperable services. In a travel booking application, services such as flight booking, hotel reservation, payment processing, and notifications work together to provide complete functionality. SOA improves flexibility, scalability, and integration in distributed systems while supporting efficient business process management.

Nov_Dec_2023

Q7) a) What is Service Oriented Architecture (SOA)? Explain the various SOA components. How does it differ from traditional software architecture?

Ans. Service Oriented Architecture (SOA)

Definition

Service Oriented Architecture (SOA) is a software architecture in which applications are developed as a collection of independent, reusable, and loosely coupled services that communicate over a network.

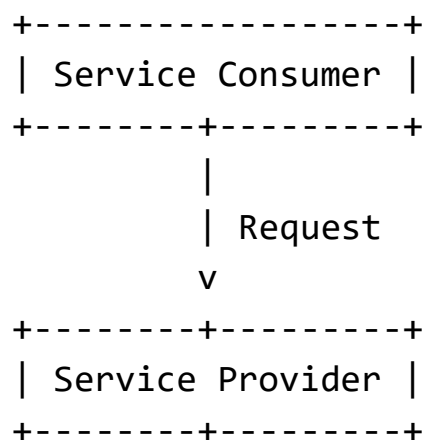
Each service performs a specific business function and can be accessed by different applications using standard communication protocols such as:

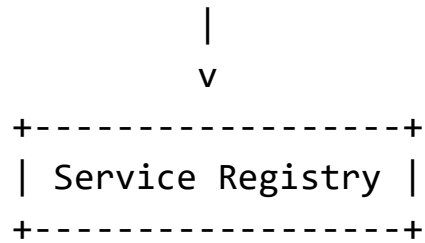
- HTTP
- SOAP
- REST
- XML
- JSON

Examples of services:

- Payment service
- Authentication service
- Booking service
- Email service

Basic SOA Architecture





Key Characteristics of SOA

1. Loose Coupling

Services are independent and interact through standard interfaces.

2. Reusability

A service can be reused in multiple applications.

3. Interoperability

Different platforms and languages can communicate.

4. Scalability

Services can be scaled independently.

5. Platform Independence

Services work across heterogeneous systems.

6. Discoverability

Services can be discovered using service registries.

Components of SOA

1. Service Provider

Provides services to clients.

2. Service Consumer

Uses the services.

3. Service Registry

Stores details of available services.

4. Service Contract

Defines communication rules and interfaces.

5. Communication Protocol

Handles data exchange between services.

Example of SOA

In an online shopping application:

- Payment service handles payments
- Inventory service manages products
- Shipping service handles delivery
- Notification service sends emails/SMS

All services work independently but cooperate through network communication.

Traditional Software Architecture

Definition

Traditional software architecture usually refers to monolithic architecture where all components of an application are tightly integrated into a single system.

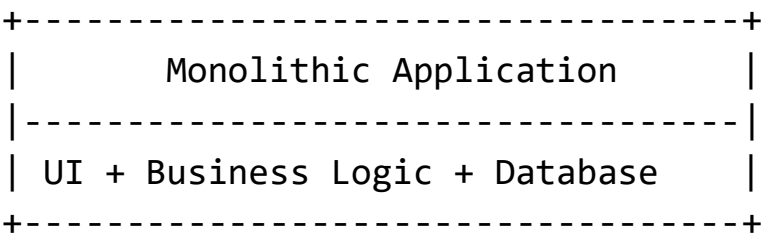
Example:

A banking application where:

- User interface
- Business logic
- Database access

are combined into one large application.

Structure of Traditional Architecture



Difference Between SOA and Traditional Software Architecture

Feature	SOA	Traditional Architecture
Architecture Style	Service-based	Monolithic
Coupling	Loosely coupled	Tightly coupled
Reusability	High	Limited
Scalability	Independent scaling	Entire application scaled together
Communication	Network-based protocols	Internal function calls
Flexibility	Highly flexible	Less flexible
Deployment	Services deployed separately	Single deployment unit
Maintenance	Easier	Difficult for large systems
Platform Support	Heterogeneous systems	Usually platform dependent
Fault Isolation	Better	Failure may affect entire system
Integration	Easy	Complex
Development	Parallel service development possible	Difficult for large teams

Advantages of SOA Over Traditional Architecture

1. Better Reusability

Services can be reused across multiple systems.

2. Improved Scalability

Only required services are scaled.

3. Easier Maintenance

Changes in one service usually do not affect others.

4. Faster Development

Teams can develop services independently.

5. Interoperability

Supports communication among different technologies.

Limitations of SOA

1. Network Dependency

Services depend on network communication.

2. Security Challenges

Distributed services increase security risks.

3. Communication Overhead

Remote service calls may increase latency.

4. Complex Service Management

Managing many distributed services is difficult.

Example Comparison

Traditional Banking System

All banking operations exist in one application:

- Account management
- Transactions
- Loans
- Notifications

Any small change may require redeploying the entire system.

SOA-Based Banking System

Separate services:

- Account service
- Transaction service
- Loan service
- Notification service

Each service can:

- Be updated independently
- Scale independently
- Be reused in other banking applications

Conclusion

Service Oriented Architecture (SOA) is a modern distributed architecture that organizes applications into reusable and loosely coupled services. Unlike traditional monolithic architecture, SOA improves flexibility, scalability, interoperability, and maintainability by enabling independent services to communicate over networks. Although SOA introduces challenges such as security and communication overhead, it is widely used in enterprise systems, cloud computing, banking, e-commerce, and distributed applications.

❖ The Future of Recent Trends

The Future of Recent Trends in Distributed Systems

Distributed systems are continuously evolving due to advances in networking, cloud computing, artificial intelligence, mobile computing, and the Internet of Things (IoT). Recent trends are shaping the future of computing toward more intelligent, scalable, secure, and highly connected systems.

Major Future Trends in Distributed Systems

1. Cloud Computing Expansion

Cloud computing will continue to dominate distributed systems.

Future developments include:

- More scalable cloud platforms
- Multi-cloud and hybrid cloud environments
- Serverless computing
- Green cloud computing

Applications:

- Online services
- Enterprise systems
- Big data analytics

Examples:

- Amazon Web Services
- Microsoft Azure
- Google Cloud

2. Edge Computing

Edge computing moves computation closer to users and devices instead of relying entirely on centralized cloud servers.

Future impact:

- Reduced latency
- Faster response time
- Better real-time processing
- Reduced network traffic

Applications:

- Autonomous vehicles

- Smart cities
- Industrial automation
- Healthcare monitoring

3. Internet of Things (IoT)

Billions of smart devices will become interconnected through distributed systems.

Future IoT developments:

- Smart homes
- Smart healthcare
- Smart agriculture
- Smart transportation

Challenges:

- Device management
- Security
- Scalability

4. Artificial Intelligence and Machine Learning Integration

AI will become deeply integrated with distributed systems.

Future capabilities:

- Intelligent resource allocation
- Predictive fault detection
- Automated system management
- Self-healing distributed systems

AI-enabled distributed systems can optimize:

- Load balancing
- Network routing
- Security monitoring

5. 5G and Future Wireless Networks

5G technology will greatly improve distributed mobile systems.

Benefits:

- Ultra-low latency
- High bandwidth
- Massive device connectivity
- Real-time communication

Future applications:

- AR/VR
- Remote surgery
- Smart factories
- Autonomous drones

6. Blockchain and Distributed Ledger Technology

Blockchain provides decentralized and secure distributed systems.

Future uses:

- Secure transactions
- Digital identity management
- Supply chain systems
- Decentralized finance (DeFi)

Advantages:

- Transparency
- Fault tolerance
- Tamper resistance

7. Distributed Big Data Processing

Massive amounts of data generated by IoT and online services will require advanced distributed processing systems.

Future technologies:

- Real-time analytics
- Distributed databases
- Stream processing

Applications:

- Social media analytics
- Scientific research
- Financial systems

8. Quantum Distributed Computing

Quantum computing may transform future distributed systems.

Potential benefits:

- Extremely fast computation
- Advanced cryptography
- Complex problem solving

Challenges:

- Hardware complexity
- Error correction
- High cost

9. Green and Energy-Efficient Computing

Future distributed systems will focus on reducing:

- Power consumption
- Carbon emissions
- Cooling requirements

Techniques:

- Energy-aware scheduling
- Efficient data centers

- Renewable energy integration

10. Cybersecurity Improvements

As distributed systems expand, security becomes increasingly important.

Future security trends:

- Zero-trust architecture
- AI-based threat detection
- Distributed authentication
- Privacy-preserving computation

11. Autonomous and Self-Managing Systems

Future distributed systems will become:

- Self-configuring
- Self-healing
- Self-optimizing

This concept is known as **autonomic computing**.

Examples:

- Automatic fault recovery
- Intelligent scaling
- Predictive maintenance

12. Wearable and Embedded Smart Devices

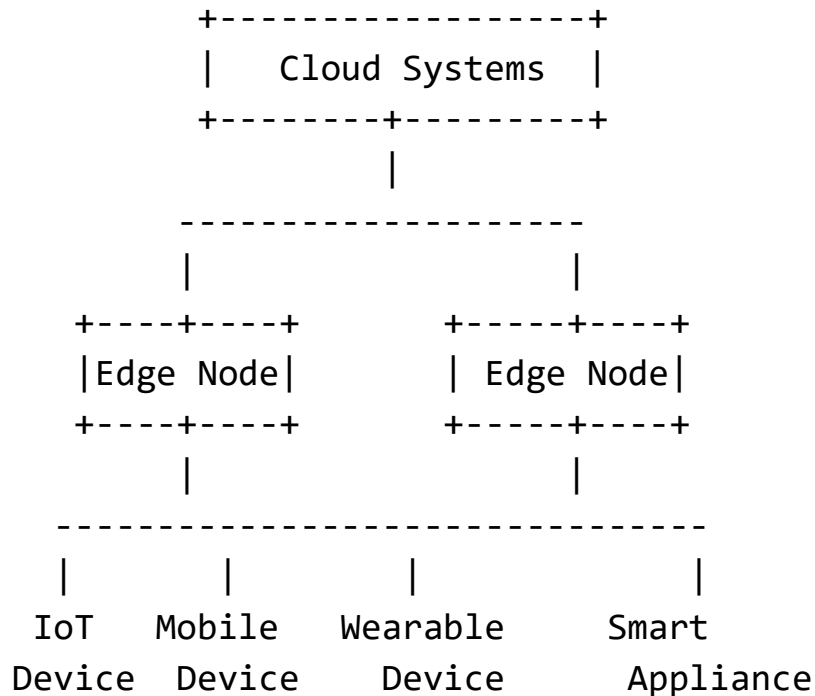
Portable and embedded devices will become more intelligent and interconnected.

Future developments:

- AI-powered wearables
- Smart medical implants
- Intelligent home appliances

- Context-aware systems

Future Architecture of Distributed Systems



Expected Characteristics of Future Distributed Systems

1. Highly Scalable

Systems will support billions of connected devices.

2. Intelligent

AI-driven automation will reduce human intervention.

3. Real-Time Processing

Low latency communication will support time-critical applications.

4. Decentralized

Blockchain and peer-to-peer systems will reduce centralized control.

5. Secure and Reliable

Advanced security mechanisms will protect distributed environments.

6. Ubiquitous Connectivity

Users and devices will remain connected everywhere.

Challenges for the Future

1. Security and Privacy

Protecting massive distributed networks from cyber attacks.

2. Data Management

Handling enormous volumes of distributed data.

3. Interoperability

Integrating heterogeneous devices and platforms.

4. Energy Consumption

Managing energy use in large-scale systems.

5. Ethical and Legal Issues

Handling AI decisions, privacy laws, and digital rights.

Real-World Future Applications

- Smart cities
- Autonomous transportation
- Remote robotic surgery
- Intelligent healthcare systems
- Industrial IoT
- Virtual and augmented reality
- Global collaborative platforms

Conclusion

The future of recent trends in distributed systems points toward intelligent, scalable, decentralized, and highly connected computing environments. Technologies such as cloud computing, edge computing, IoT, AI, blockchain, 5G, and quantum computing will continue transforming how distributed

systems are designed and used. Although challenges related to security, scalability, interoperability, and energy efficiency remain, future distributed systems will provide faster, smarter, and more reliable services for industries and everyday life.

➤ **Tools for Distributed System Monitoring**

❖ **Prometheus**

❖ **Zabbix**

❖ **Nagios**

May_Jun_2024

Q8) a) Compare the following tools for Distributed System Monitoring : Prometheus, Zabbix, Nagios.

Ans. Comparison of Distributed System Monitoring Tools: Prometheus, Zabbix, and Nagios

Distributed systems require continuous monitoring to ensure:

- Availability
- Performance
- Fault detection
- Scalability
- Resource utilization

Three widely used monitoring tools are:

- Prometheus
- Zabbix
- Nagios

1. Prometheus

Definition

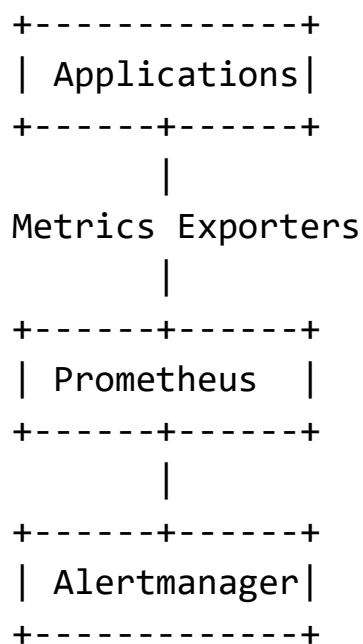
Prometheus is an open-source monitoring and alerting toolkit mainly designed for cloud-native and containerized distributed systems.

Originally developed by SoundCloud and now maintained by the Cloud Native Computing Foundation.

Features

- Time-series database
- Pull-based data collection
- Powerful query language (PromQL)
- Real-time monitoring
- Kubernetes integration
- Dynamic service discovery
- Alert management

Architecture



Advantages

- Excellent for cloud and microservices
- Highly scalable
- Strong Kubernetes support
- Fast querying and visualization

Disadvantages

- Limited long-term storage by default
- Less suitable for traditional network monitoring
- Complex query language for beginners

Best Use Cases

- Cloud-native applications
- Kubernetes clusters
- Container monitoring
- DevOps environments

2. Zabbix

Definition

Zabbix is an enterprise-level open-source monitoring platform used for monitoring:

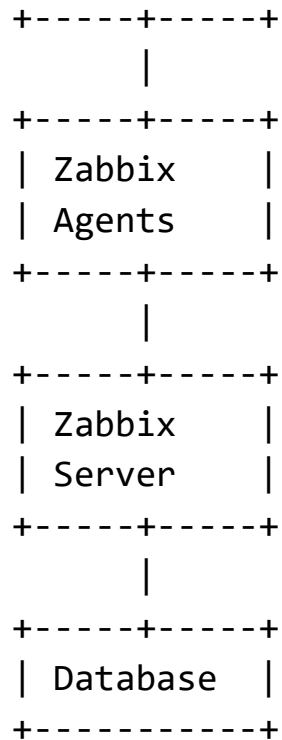
- Servers
- Networks
- Applications
- Databases
- Cloud services

Features

- Agent-based and agentless monitoring
- Real-time alerts
- Web-based dashboard
- Auto-discovery
- Distributed monitoring
- Visualization and reporting

Architecture

```
+-----+  
| Devices |
```



Advantages

- Easy graphical interface
- Comprehensive monitoring
- Strong alerting features
- Good visualization support
- Supports large enterprise systems

Disadvantages

- Database-heavy architecture
- Can become complex at large scale
- Higher resource usage

Best Use Cases

- Enterprise infrastructure monitoring
- Server and network monitoring
- Data center monitoring

3. Nagios

Definition

Nagios is one of the oldest and most widely used open-source infrastructure monitoring systems.

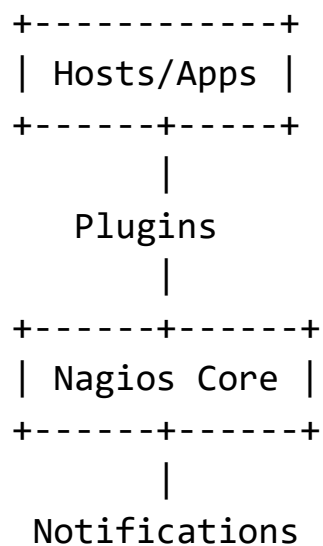
It focuses mainly on:

- Network monitoring
- Server monitoring
- Service availability

Features

- Plugin-based architecture
- Alerting and notifications
- Host and service monitoring
- Event handling
- Extensible framework

Architecture



Advantages

- Very stable and reliable

- Large plugin ecosystem
- Good fault detection
- Lightweight core system

Disadvantages

- Old user interface
- Manual configuration can be difficult
- Limited scalability compared to modern tools
- Less suitable for dynamic cloud systems

Best Use Cases

- Traditional server monitoring
- Network infrastructure monitoring
- Small to medium distributed systems

Comparison Table

Feature	Prometheus	Zabbix	Nagios
Primary Focus	Cloud-native monitoring	Enterprise infrastructure monitoring	Network/server monitoring
Monitoring Type	Time-series metrics	Full-stack monitoring	Host/service monitoring
Data Collection	Pull model	Agent/agentless	Plugins/agents
Scalability	Very high	High	Moderate
Cloud Support	Excellent	Good	Limited
Kubernetes Support	Excellent	Moderate	Limited
Visualization	Basic (with Grafana)	Built-in dashboards	Limited
Alerting	Strong	Strong	Strong
Ease of Configuration	Moderate	Easy to Moderate	Difficult
UI Experience	Modern	Modern	Older UI
Database Requirement	Built-in TSDB	External DB required	Minimal
Dynamic Discovery	Excellent	Good	Limited
Best For	DevOps, Containers	Enterprises	Traditional IT systems

Prometheus vs Zabbix vs Nagios

Prometheus

Best for:

- Modern cloud-native distributed systems
- Microservices
- Kubernetes monitoring

Strong in:

- Scalability

- Real-time metrics
- Dynamic environments

Zabbix

Best for:

- Enterprise IT infrastructure
- Hybrid environments
- Comprehensive monitoring

Strong in:

- Visualization
- Integrated monitoring
- Ease of use

Nagios

Best for:

- Traditional server/network monitoring
- Smaller infrastructures

Strong in:

- Stability
- Plugin ecosystem
- Fault monitoring

Integration with Visualization Tools

Prometheus and Zabbix are commonly integrated with:

- Grafana

Grafana provides:

- Dashboards
- Real-time graphs
- Analytics visualization

Conclusion

Prometheus, Zabbix, and Nagios are powerful monitoring tools used in distributed systems, but they are designed for different environments and requirements. Prometheus is ideal for cloud-native and containerized systems, Zabbix is suitable for enterprise-wide infrastructure monitoring, and Nagios is effective for traditional server and network monitoring. The selection of a monitoring tool depends on scalability needs, infrastructure type, cloud integration, and monitoring complexity.

Nov_Dec_2024

Q7) b) Explain in brief, the key features of Zabbix including auto-discovery, triggers, or dashboards.

Ans. Key Features of Zabbix

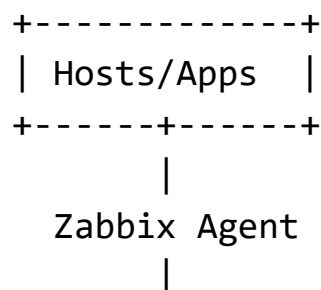
Introduction

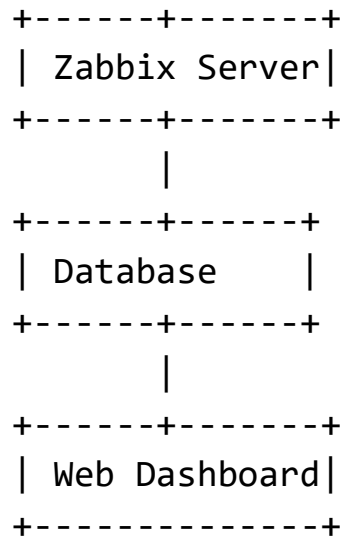
Zabbix is an open-source enterprise monitoring tool used for monitoring:

- Servers
- Networks
- Applications
- Databases
- Cloud infrastructure
- Distributed systems

It provides real-time monitoring, alerting, visualization, and automated management of IT infrastructure.

Architecture of Zabbix





Key Features of Zabbix

1. Auto-Discovery

Definition

Auto-discovery automatically detects devices, servers, services, and network components connected to the system.

It reduces manual configuration work in large distributed environments.

How It Works

Zabbix scans networks and identifies:

- New hosts
- Network devices
- Services
- IP addresses

It can automatically:

- Add discovered devices
- Assign templates
- Start monitoring

Example

Suppose a new Linux server is added to a distributed cloud environment.

Zabbix:

1. Detects the server
2. Registers it automatically
3. Applies monitoring templates
4. Starts collecting metrics

Benefits

- Reduces administrative effort
- Supports dynamic infrastructures
- Useful for cloud and virtualized systems
- Faster deployment

2. Triggers

Definition

Triggers are logical conditions used to detect problems or unusual situations based on collected metrics.

A trigger changes system status when a condition becomes true.

Examples of Trigger Conditions

- CPU usage > 90%
- Disk space < 10%
- Server unavailable
- High network latency

Working of Triggers

Metric Collection --> Trigger Evaluation --> Alert Generation

Example

Trigger expression:

CPU Usage > 90% for 5 minutes

If the condition is satisfied:

- Zabbix generates an alert
- Notifications are sent to administrators

Trigger Severity Levels

Zabbix supports multiple severity levels:

- Information
- Warning
- Average
- High
- Disaster

Benefits

- Automatic fault detection
- Real-time alerts
- Faster issue resolution
- Preventive monitoring

3. Dashboards

Definition

Dashboards provide graphical visualization of monitored systems and metrics in real time.

They help administrators quickly understand infrastructure status.

Dashboard Components

Dashboards can display:

- CPU utilization
- Memory usage
- Network traffic

- Availability status
- Problem summaries
- Graphs and charts

Example Dashboard

+-----+		
CPU Usage	Network Traffic	
Memory Usage	Active Alerts	
Server Status	Disk Space	
+-----+		

Features of Dashboards

Customizable Layouts

Users can create personalized dashboards.

Real-Time Visualization

Data updates continuously.

Multi-Screen Support

Useful for network operation centers (NOCs).

Interactive Graphs

Allows detailed metric analysis.

Benefits

- Simplifies monitoring
- Improves visibility
- Helps in quick decision-making
- Supports centralized monitoring

4. Alerting and Notification

Definition

Zabbix sends alerts when triggers detect problems.

Notification Methods

- Email
- SMS
- Slack
- Telegram
- Webhooks

Example

If a database server crashes:

- Zabbix immediately sends notifications to administrators.

5. Distributed Monitoring

Definition

Zabbix supports monitoring across geographically distributed systems.

Features

- Proxy servers
- Remote monitoring
- Centralized management

Example

A company can monitor:

- Branch offices
- Data centers
- Cloud servers

from one central Zabbix server.

6. Agent-Based and Agentless Monitoring

Agent-Based

A Zabbix agent installed on hosts collects detailed system metrics.

Agentless

Uses protocols such as:

- SNMP
- SSH
- IPMI
- HTTP

without installing agents.

7. Data Collection and Metrics

Zabbix collects:

- CPU usage
- RAM usage
- Disk utilization
- Network traffic
- Application performance

in real time.

8. Visualization and Reporting

Zabbix provides:

- Graphs
- Maps
- Screens
- Historical reports

for performance analysis and troubleshooting.

9. Template-Based Monitoring

Definition

Templates are predefined monitoring configurations.

Benefits

- Reusability
- Standardization
- Easy deployment

Example

A Linux template can monitor:

- CPU
- Memory
- Disk
- Network

for all Linux servers automatically.

10. Scalability

Zabbix can monitor:

- Thousands of hosts
- Millions of metrics
- Large enterprise infrastructures

Advantages of Zabbix

- Open source
- Real-time monitoring
- Flexible alerting
- Centralized monitoring
- Strong visualization support
- Enterprise scalability

Limitations of Zabbix

- Complex setup for large systems
- Database-intensive

- Learning curve for advanced configurations

Applications of Zabbix

- Data center monitoring
- Cloud infrastructure monitoring
- Network monitoring
- Server monitoring
- IoT monitoring
- Distributed systems management

Conclusion

Zabbix is a powerful enterprise monitoring solution for distributed systems and IT infrastructure. Its key features such as auto-discovery, triggers, dashboards, alerting, distributed monitoring, and visualization help organizations efficiently monitor and manage complex environments. Zabbix improves system reliability, fault detection, and operational efficiency through real-time monitoring and automated management capabilities.

Q8) a) Explain in brief, the following Distributed System monitoring tools: i) Nagios ii) Zabbix

Ans. Nagios Distributed System Monitoring Tool

Introduction

Nagios is one of the most popular open-source monitoring tools used for monitoring distributed systems, networks, servers, applications, and IT infrastructure.

It helps system administrators:

- Detect failures
- Monitor performance
- Generate alerts
- Ensure high availability of distributed systems

Nagios is widely used in:

- Data centers
- Cloud systems
- Enterprise networks
- Distributed computing environments

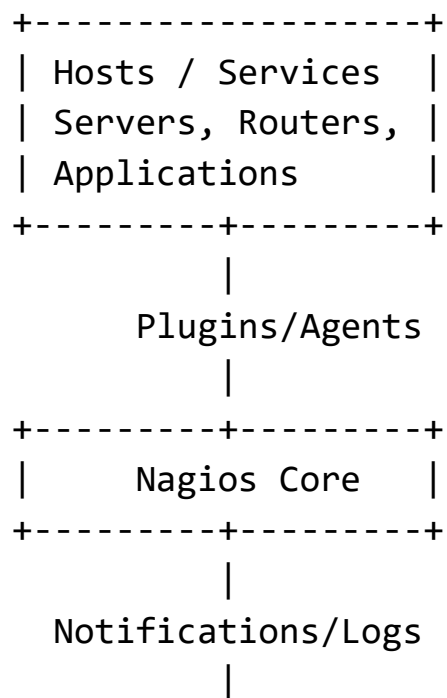
Definition

Nagios is a host and service monitoring tool that continuously checks the status and health of systems, applications, and network devices in a distributed environment.

It provides:

- Real-time monitoring
- Fault detection
- Notification services
- Performance analysis

Architecture of Nagios



```
+-----+-----+
| Admin Dashboard |
+-----+-----+
```

Components of Nagios

1. Nagios Core

Definition

The main monitoring engine responsible for:

- Scheduling checks
- Processing results
- Triggering alerts
- Managing events

Functions

- Executes monitoring tasks
- Detects failures
- Maintains status information
- Generates notifications

2. Plugins

Definition

Plugins are external programs used to collect monitoring information.

Nagios uses plugins to monitor:

- CPU usage
- Memory
- Disk space
- Network services
- Applications

Examples

- HTTP plugin
- Ping plugin
- SSH plugin
- Database plugin

3. Hosts

Definition

Hosts are systems or devices being monitored.

Examples:

- Servers
- Routers
- Switches
- Cloud instances

4. Services

Definition

Services are specific functionalities monitored on hosts.

Examples:

- Web server status
- Database availability
- CPU utilization
- Mail service

5. NRPE (Nagios Remote Plugin Executor)

Definition

NRPE allows Nagios to execute plugins remotely on distributed hosts.

Purpose

Used for:

- Remote Linux monitoring
- Distributed server monitoring

6. Notification System

Nagios sends alerts when problems occur.

Notification Methods

- Email
- SMS
- Slack
- Webhooks

Example

If a web server goes down:

- Nagios immediately sends an email alert to administrators.

Working of Nagios

Step 1: Monitoring Setup

Hosts and services are configured in Nagios.

Step 2: Periodic Checks

Nagios periodically checks:

- Host availability
- Service performance

using plugins.

Step 3: Status Evaluation

Nagios determines whether the system is:

- OK
- Warning
- Critical
- Unknown

Step 4: Alert Generation

If a problem is detected:

- Alerts are generated
- Notifications are sent

Step 5: Recovery Detection

When the issue is resolved:

- Nagios updates system status
- Recovery notifications are sent

Monitoring States in Nagios

Host States

- UP
- DOWN
- UNREACHABLE

Service States

- OK
- WARNING
- CRITICAL
- UNKNOWN

Features of Nagios

1. Distributed Monitoring

Nagios supports monitoring geographically distributed systems using:

- Remote agents
- NRPE
- Distributed servers

2. Plugin-Based Architecture

Thousands of plugins are available for monitoring:

- Servers
- Databases
- Networks
- Applications

3. Real-Time Alerting

Provides immediate alerts for failures and performance issues.

4. Scalability

Can monitor:

- Hundreds or thousands of devices
- with proper configuration.

5. Event Handling

Nagios can automatically execute scripts when events occur.

Example:

- Restarting failed services automatically.

6. Web Interface

Provides a dashboard showing:

- System health
- Alerts
- Performance data

7. Logging and Reporting

Nagios stores:

- Historical logs
- Availability reports
- Performance statistics

Example: Nagios in Distributed System Monitoring

Suppose an organization has:

- Multiple servers
- Cloud instances
- Databases
- Network devices

distributed across different locations.

Nagios:

1. Continuously checks all devices
2. Detects failures
3. Sends alerts
4. Displays status on dashboards
5. Maintains logs and reports

Advantages of Nagios

- Open-source and free
- Stable and reliable

- Large plugin ecosystem
- Strong alerting capabilities
- Flexible monitoring
- Supports distributed systems

Limitations of Nagios

- Complex manual configuration
- Older user interface
- Limited built-in visualization
- Less suitable for dynamic cloud-native systems
- Scalability challenges in very large infrastructures

Applications of Nagios

- Server monitoring
- Network monitoring
- Cloud monitoring
- Data center management
- Distributed system monitoring
- IT infrastructure management

Nagios vs Modern Monitoring Tools

Feature	Nagios	Prometheus
Architecture	Plugin-based	Time-series based
Best For	Traditional infrastructure	Cloud-native systems
Data Collection	Active checks	Pull-based metrics
Kubernetes Support	Limited	Excellent
Visualization	Basic	Better with Grafana

Conclusion

Nagios is a powerful and reliable distributed system monitoring tool widely used for monitoring servers, networks, applications, and IT infrastructure. Its plugin-based architecture, real-time alerting, distributed monitoring support, and fault detection capabilities make it highly effective for traditional distributed environments. Although newer tools like Prometheus are more suitable for cloud-native systems, Nagios remains an important monitoring solution for enterprise infrastructure management.

May_Jun_2025

Q7) b) Explain in brief, the key features of Prometheus including data model, query language, or alerting rules.

Ans. Key Features of Prometheus

Introduction

Prometheus is an open-source monitoring and alerting toolkit designed for modern distributed systems, cloud-native applications, and microservices architectures.

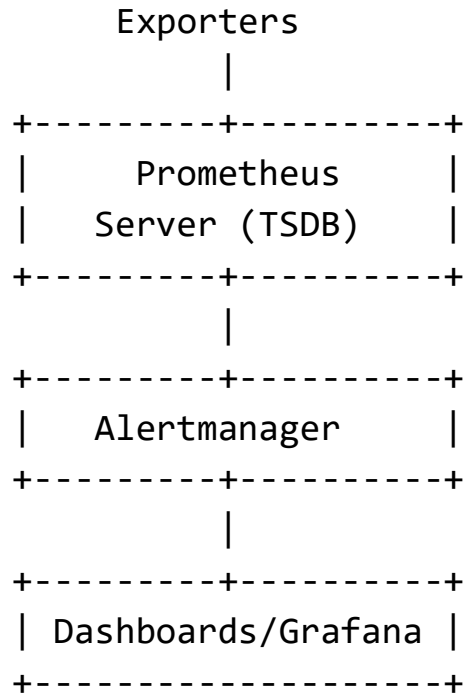
Originally developed by SoundCloud and now maintained by the Cloud Native Computing Foundation.

Prometheus is widely used with:

- Kubernetes
- Containers
- Cloud platforms
- Distributed applications

Architecture of Prometheus

```
+-----+
| Applications/Nodes |
+-----+-----+
      |
```



Key Features of Prometheus

1. Data Model

Definition

Prometheus uses a multidimensional time-series data model.

Data is stored as:

- Metrics
- Timestamp
- Key-value labels

Structure of Data

Each metric contains:

- Metric name
- Labels
- Timestamp

- Value

Example Metric

`cpu_usage{host="server1", region="india"} 85`

Where:

- `cpu_usage` → metric name
- `host, region` → labels
- `85` → metric value

Time-Series Data

Prometheus continuously records metric values over time.

This helps:

- Trend analysis
- Performance monitoring
- Capacity planning

Advantages of the Data Model

- Flexible querying
- Efficient storage
- Multi-dimensional analysis
- Easy aggregation and filtering

2. Pull-Based Data Collection

Definition

Prometheus collects metrics by periodically pulling data from targets.

Working

Applications expose metrics using HTTP endpoints such as:

`/metrics`

Prometheus scrapes these endpoints at regular intervals.

Advantages

- Simpler architecture
- Better reliability
- Easy service discovery
- Centralized monitoring

3. Exporters

Definition

Exporters expose metrics from systems and applications.

Examples of Exporters

- Node Exporter → system metrics
- MySQL Exporter → database metrics
- Blackbox Exporter → network probing

Example

Node Exporter provides:

- CPU usage
- Memory utilization
- Disk usage
- Network statistics

4. Query Language (PromQL)

Definition

Prometheus provides a powerful query language called PromQL (Prometheus Query Language).

It is used for:

- Data analysis
- Aggregation
- Filtering

- Alerting

Examples of PromQL Queries

CPU Usage Query

```
cpu_usage > 80
```

Average Memory Usage

```
avg(memory_usage)
```

Rate of HTTP Requests

```
rate(http_requests_total[5m])
```

Features of PromQL

- Aggregation functions
- Mathematical operations
- Filtering using labels
- Time-based calculations
- Statistical analysis

Advantages of PromQL

- Very flexible
- Real-time analytics
- Easy metric correlation
- Supports complex monitoring scenarios

5. Alerting Rules

Definition

Alerting rules define conditions under which alerts should be generated.

Example Alert Rule

IF CPU usage > 90% FOR 5 minutes
THEN generate alert

Working of Alerting

Metric Collection



PromQL Evaluation



Alert Rule Triggered



Alertmanager



Notification Sent

Alertmanager

Prometheus uses Alertmanager to:

- Manage alerts
- Group alerts
- Silence duplicate alerts
- Send notifications

Notification Methods

- Email
- Slack
- PagerDuty
- Webhooks

Benefits of Alerting Rules

- Real-time fault detection
- Preventive monitoring

- Faster incident response
- Automated alert management

6. Service Discovery

Definition

Prometheus automatically discovers monitoring targets.

Supported Environments

- Kubernetes
- Docker
- Cloud platforms
- Virtual machines

Benefits

- Dynamic monitoring
- Easy scaling
- Reduced manual configuration

7. Time-Series Database (TSDB)

Definition

Prometheus includes a built-in time-series database.

Features

- Efficient compression
- Fast querying
- Local storage
- High-performance metric storage

8. Visualization Support

Prometheus integrates with:

- Grafana

Grafana provides:

- Dashboards
- Graphs
- Analytics visualization

9. Kubernetes Integration

Prometheus is highly integrated with:

- Kubernetes clusters
- Containers
- Microservices

It is widely used in DevOps and cloud-native systems.

10. Scalability

Prometheus supports:

- Large-scale distributed monitoring
- Thousands of targets
- High-frequency metric collection

Advantages of Prometheus

- Open-source
- Powerful query language
- Excellent Kubernetes support
- Real-time monitoring
- Flexible alerting
- Cloud-native architecture

Limitations of Prometheus

- Limited long-term storage
- Pull model may not fit all scenarios

- Complex PromQL for beginners
- Less suitable for log monitoring

Applications of Prometheus

- Kubernetes monitoring
- Cloud infrastructure monitoring
- Microservices monitoring
- Container monitoring
- DevOps observability
- Distributed systems management

Example Use Case

In a cloud-based e-commerce system:

Prometheus monitors:

- Web servers
- Databases
- APIs
- Containers

It:

- Collects metrics
- Detects failures
- Sends alerts
- Displays dashboards

in real time.

Conclusion

Prometheus is a modern and powerful monitoring system designed for distributed and cloud-native environments. Its key features such as multidimensional data model, PromQL query language, alerting rules, service discovery, exporters, and time-series database make it highly effective for

monitoring large-scale distributed systems. Prometheus plays a major role in DevOps, Kubernetes monitoring, and real-time observability of modern applications.

◆ **Case Studies**

❖ **Mach**

❖ **Chorus**

Nov_Dec_2023

Q8) b) Provide an overview of Mach and CHORUS microkernels. How are memory management techniques used to avoid physical copying of data in Mach and CHORUS?

Ans. Overview of Mach and CHORUS Microkernels

[Introduction to Microkernels](#)

A microkernel is a minimal operating system kernel that provides only essential services such as:

- Process management
- Memory management
- Inter-process communication (IPC)
- Basic scheduling

Other services like:

- File systems
- Device drivers
- Networking

run in user space as separate servers.

Advantages:

- Modularity
- Reliability
- Portability
- Better fault isolation

Two important microkernel-based distributed operating systems are:

- Mach
- CHORUS

1. Mach Microkernel

Definition

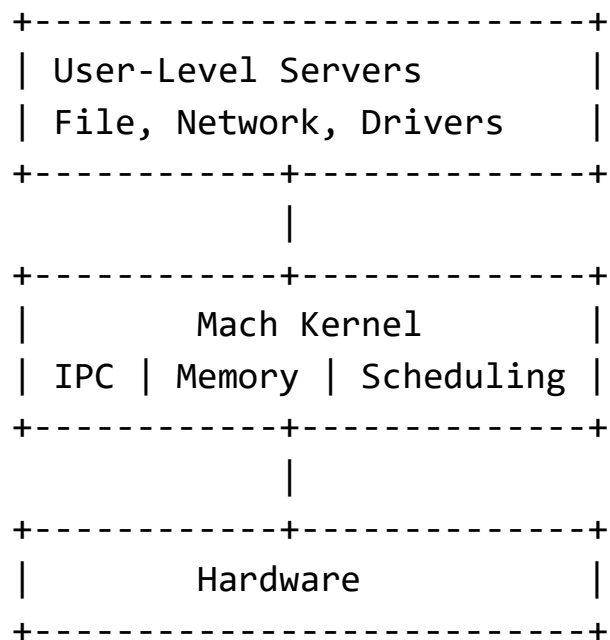
Mach is a microkernel developed at Carnegie Mellon University to support:

- Distributed computing
- Parallel processing
- Multiprocessor systems

Mach forms the basis for systems such as:

- macOS kernel components
- NeXTSTEP

Architecture of Mach



Key Features of Mach

1. Message-Based IPC

Communication occurs using message passing.

2. Virtual Memory Support

Mach provides advanced memory management.

3. Port-Based Communication

Processes communicate through ports.

4. Multiprocessor Support

Efficient for parallel computing systems.

5. Distributed System Support

Supports transparent distributed communication.

2. CHORUS Microkernel

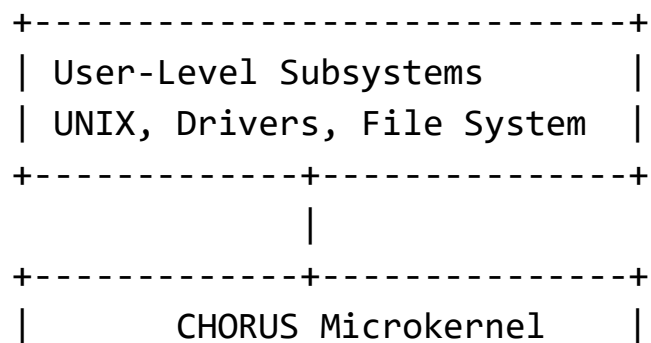
Definition

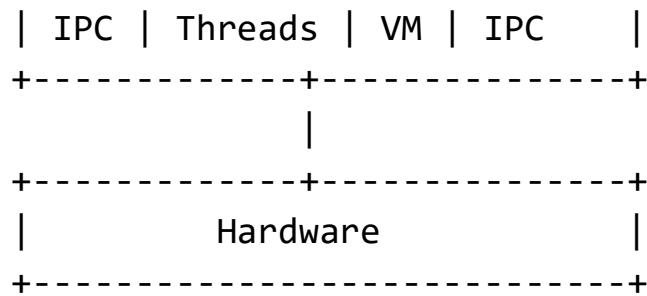
CHORUS is a distributed microkernel operating system designed for:

- Distributed applications
- Real-time systems
- Embedded systems

It was developed by Chorus Systèmes.

Architecture of CHORUS





Key Features of CHORUS

1. Object-Oriented Design

Resources are represented as objects.

2. Distributed Processing

Supports communication among distributed nodes.

3. Real-Time Support

Suitable for embedded and real-time applications.

4. Lightweight Processes

Efficient thread management.

5. Modular Architecture

System services execute outside the kernel.

Comparison of Mach and CHORUS

Feature	Mach	CHORUS
Developed By	Carnegie Mellon University	Chorus Systèmes
Main Focus	Distributed & parallel computing	Distributed & real-time systems
IPC Mechanism	Message passing using ports	Message-based object communication
Memory Management	Advanced virtual memory	Distributed virtual memory
Real-Time Support	Limited	Strong
Architecture	Microkernel	Microkernel
Object-Oriented	Partial	Strongly object-oriented

Avoiding Physical Copying of Data

In distributed systems, copying data physically between processes or machines:

- Increases overhead
- Reduces performance
- Wastes memory bandwidth

Both Mach and CHORUS use advanced memory management techniques to avoid unnecessary physical copying of data.

Memory Management in Mach

Copy-on-Write (COW)

Definition

Mach uses a technique called **Copy-on-Write (COW)** to avoid immediate physical copying.

Working of Copy-on-Write

When a process shares data:

1. Memory pages are shared virtually.
2. No physical copy is created initially.
3. Actual copying occurs only when one process modifies the data.

Diagram

```
Process A ---- \
                  \ Shared Memory Page
Process B ---- /
```

Initially:

- Both processes point to the same physical page.

If Process B modifies data:

- A new copy of the page is created.

Advantages

- Reduces memory usage
- Improves performance
- Faster IPC
- Efficient process creation

Memory Mapping in Mach

Mach also uses:

- Memory-mapped communication
- Virtual memory remapping

Instead of copying messages:

- Kernel remaps memory pages between processes.

This avoids:

- Data duplication
- CPU overhead

Example in Mach IPC

Instead of:

Copy Sender Buffer → Kernel → Receiver Buffer

Mach performs:

Remap Memory Pages

Result:

- Faster communication
- Reduced physical copying

Memory Management in CHORUS

CHORUS also uses virtual memory techniques to minimize copying.

Distributed Virtual Memory (DVM)

Definition

CHORUS provides distributed virtual memory where processes can share memory objects across nodes.

Working

- Memory objects are mapped into process address spaces.
- Data pages are shared virtually.
- Physical copying occurs only when necessary.

Page Remapping

Instead of copying large data blocks:

- CHORUS remaps pages between processes.

This technique:

- Transfers ownership of memory pages
- Avoids duplicate copying

Shared Memory Objects

Processes communicate using shared objects stored in virtual memory.

Example:

Process A <---- Shared Object ----> Process B

Both access the same memory object logically.

Advantages in CHORUS

- Reduced communication overhead
- Faster distributed IPC
- Better memory efficiency
- Improved distributed performance

Comparison of Copy Avoidance Techniques

Technique	Mach	CHORUS
Copy-on-Write	Yes	Yes
Memory Mapping	Yes	Yes
Page Remapping	Yes	Yes
Distributed Virtual Memory	Limited	Strong
Shared Memory Objects	Supported	Extensively used

Benefits of Avoiding Physical Copying

- Improved performance
- Reduced CPU overhead
- Lower memory usage
- Faster IPC
- Better scalability
- Efficient distributed communication

Conclusion

Mach and CHORUS are important microkernel-based distributed operating systems that emphasize modularity, IPC, and virtual memory management. Mach focuses mainly on distributed and parallel processing, while CHORUS provides strong support for distributed and real-time systems. Both systems use advanced memory management techniques such as copy-on-write, memory mapping, page remapping, and distributed virtual memory to avoid unnecessary physical copying of data, thereby improving performance and efficiency in distributed environments.